

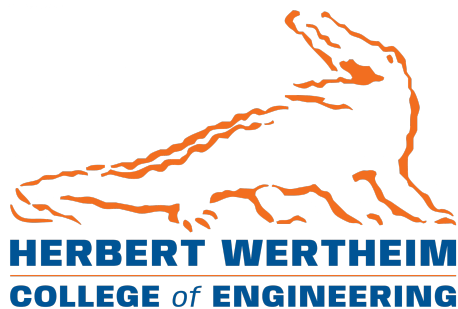
# T2: Beta Test Plan

## EEL4908 CpE Design 2

Hardened RISC-V Exam Computer

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## 1 Scope

**Beta objective.** Two boards accept a “start SEB” command from the server, and SEB launches on both.

Repository: <https://github.com/cannon-spencer/riscv-exam-computer>. Commit tested: 027d47894359fdc4bdf7c9e229fe81d56dff1a36

**In scope.** Control server, board agent, start and stop scripts, SEB session on the board. *[TODO]* (add or remove items)

**Out of scope.** *[TODO]*

**Notation.** *[TODO]* marks something for the team to fill in or confirm; where the blank is not obvious, a short note after the colon says what to write. Nothing marked TODO has been measured or decided.

## 2 Alpha test results

Describe the results of the alpha testing, how they changed the project, and how they influenced this plan. In the table below, give each test a result (Pass, Fail or Not run) and note the evidence.

### 2.1 Results against the T1 plan

| T1 ID  | Result        | Evidence and notes |
|--------|---------------|--------------------|
| HW-01  | <i>[TODO]</i> | <i>[TODO]</i>      |
| HW-02  | <i>[TODO]</i> | <i>[TODO]</i>      |
| HW-03  | <i>[TODO]</i> | <i>[TODO]</i>      |
| SW-01  | <i>[TODO]</i> | <i>[TODO]</i>      |
| INT-01 | <i>[TODO]</i> | <i>[TODO]</i>      |
| INT-02 | <i>[TODO]</i> | <i>[TODO]</i>      |
| SW-02  | <i>[TODO]</i> | <i>[TODO]</i>      |

Table 1: Alpha results.

### 2.2 How alpha testing changed the project

*[TODO: for each alpha finding, say what you saw and what you changed]*

### 2.3 How alpha testing shaped this plan

*[TODO: for each alpha finding, give the beta test ID that checks it]*

## 3 Expected behavior

This section describes behavior as implemented in the source at the tested commit.

### 3.1 System overview

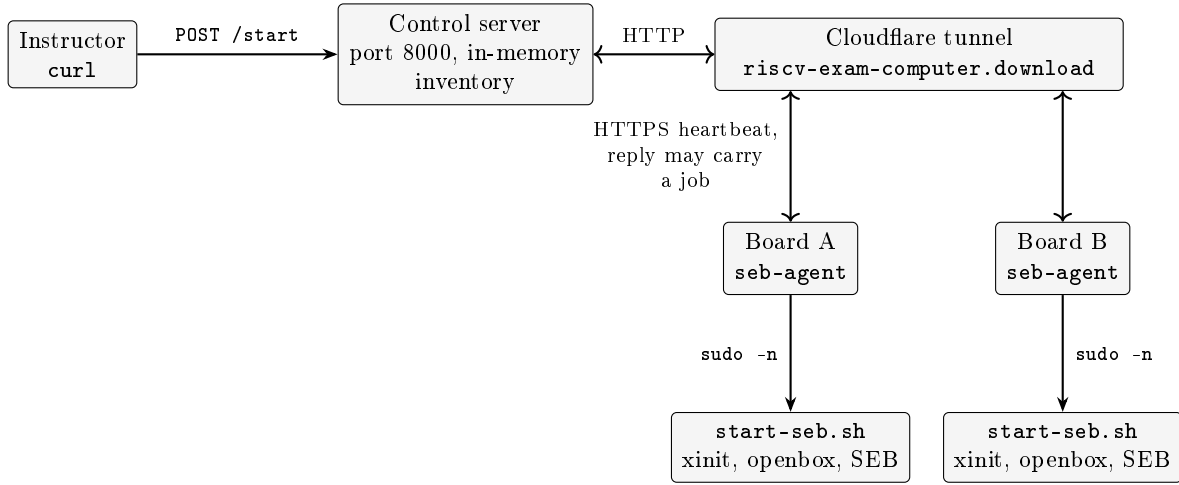


Figure 1: Control path. Boards poll the server; the server never connects to a board.

### 3.2 Control server

The server (`exam-env/server/src/main.rs`) listens on port 8000 and holds, per host name, the last heartbeat time, the last reported state and at most one pending job (`start` or `stop`).

| Route           | Expected behavior                                                                                                                                                                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| POST /heartbeat | Body <code>{"host":..., "state":...}</code> . Records the host, time and state. Replies <code>{"ok":true, "job":...}</code> where <code>job</code> is <code>"start"</code> , <code>"stop"</code> or <code>null</code> . Missing <code>host</code> : 422. Invalid JSON: 400. No JSON content type: 415. |
| GET /machines   | Lists every host seen since the server started, with <code>state</code> , <code>pending</code> and <code>seen_secs_ago</code> , sorted by host.                                                                                                                                                        |
| POST /start     | Sets pending <code>start</code> on every host whose last heartbeat is 15 s old or less, and returns the sorted list <code>queued</code> . Returns <code>"queued": []</code> when no host is online.                                                                                                    |
| POST /stop      | Same as <code>/start</code> with job <code>stop</code> .                                                                                                                                                                                                                                               |

Table 2: Server routes.

Job delivery rules (checked by SRV-01 to SRV-09):

- A job is handed out in the reply to the board's next heartbeat and then cleared.
- A pending `start` is not delivered while the board reports `running`; it stays pending and is delivered the first time the board reports another state.
- A pending `stop` is delivered regardless of reported state.
- There is one pending slot per board, so a later command replaces an earlier undelivered one.
- Inventory is held in memory only.

### 3.3 Board agent

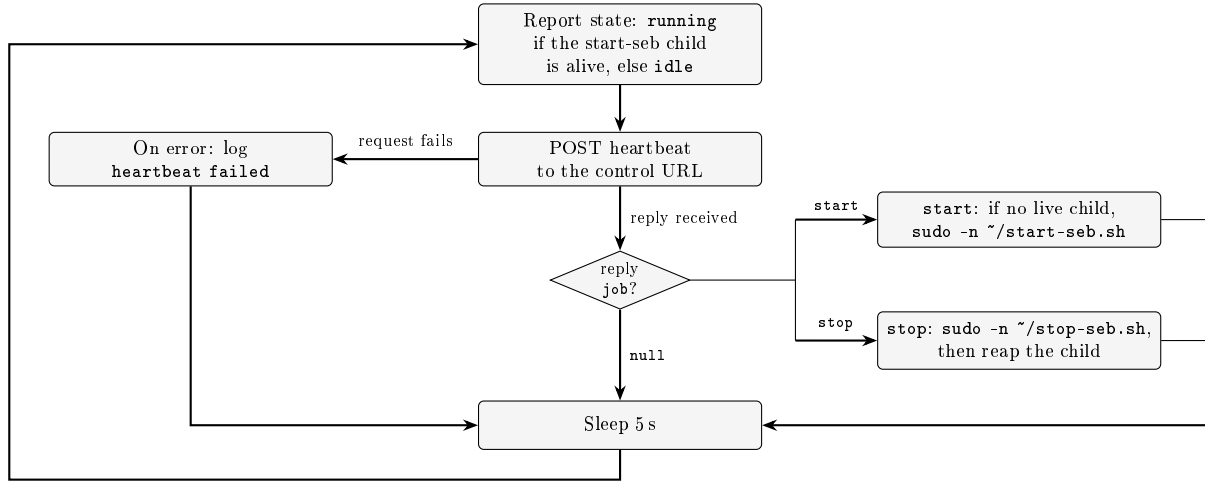


Figure 2: Agent loop (*exam-env/seb-agent/src/main.rs*).

- The agent sleeps 5 s after each POST, so the observed period is 5 s plus the request. It reports **running** while the `sudo/xinit` child from `start-seb.sh` is alive, otherwise **idle**.
- The reported host name is the `HOSTNAME` environment variable, set by the `systemd` unit from the `%H` specifier.
- The control URL is compiled into the agent: `https://riscv-exam-computer.download/heartbeat`.
- The agent opens no listening port.
- Sudo for the agent is limited to `/home/orangepi/start-seb.sh` and `/home/orangepi/stop-seb.sh`.

### 3.4 Board session

**Start.** `start-seb.sh` runs as root and execs `xinit` on display `:1`, virtual terminal 3, starting `openbox` and then `/home/orangepi/safe-exam-browser--windowed` with a local start file (paths hardcoded for the `orangepi` account).

**Stop.** `stop-seb.sh` kills the SEB and `xinit` processes and X1 socket holders, rebinds the frame-buffer console, restores keyboard mode and switches to virtual terminal 1.

### 3.5 Quantified requirements

| ID   | Requirement                                                            | Value                                |
|------|------------------------------------------------------------------------|--------------------------------------|
| R-T1 | Heartbeat sleep after each request                                     | 5 s (observed period is sleep + RTT) |
| R-T2 | A board counts as online while its last heartbeat is no older than     | 15 s (code)                          |
| R-T3 | Time from POST /start to SEB running on every online board, in seconds | [TODO]                               |
| R-T4 | Maximum difference between the two boards' start times, in seconds     | [TODO]                               |
| R-T5 | Time from POST /stop to console restored, in seconds                   | [TODO]                               |
| R-P1 | Canvas page load time, in seconds                                      | [TODO]                               |
| R-P2 | Click-to-response latency in Canvas, in seconds                        | [TODO]                               |
| R-P3 | Maximum board temperature during a session, in °C                      | [TODO]                               |

Table 3: Quantified requirements. Rows marked *TODO* are to be set by the team.

### 3.6 Not implemented in the tested commit

- The server code has no authentication on any route.
- /start and /stop address all online boards; there is no per-board targeting.
- The server does not distribute exam configuration; a later change is a pushed .seb.
- Secure start (the exam opening only under SEB): [TODO: describe what should happen, and how it would be tested]
- Other: [TODO]

## 4 Test procedures

### 4.1 Equipment and conventions

[TODO: list the boards, displays, laptop, network and tools you actually use] Record the commit and board names for every run in Table 10.

### 4.2 Server (no hardware)

Start a local server on :8000 (cargo run in exam-env/server, or Docker without the tunnel). Do not use ./scripts/deploy-server.sh or the public URL for these tests: that path is the live tunnel, and live boards would consume jobs. Drive SRV-01 to SRV-09 with curl to http://127.0.0.1:8000. SRV-09 needs a 16 s wait after the last heartbeat.

| ID     | Procedure and pass criterion                                                                                                                                          |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SRV-01 | <b>Heartbeat.</b> POST a heartbeat for a new host. <b>Pass:</b> reply contains "ok":true and "job":null.                                                              |
| SRV-02 | <b>Inventory.</b> Heartbeat two hosts, then GET /machines. <b>Pass:</b> both hosts listed with state and seen_secs_ago.                                               |
| SRV-03 | <b>Broadcast start.</b> POST /start with both hosts online. <b>Pass:</b> queued lists both; the next heartbeat from a host returns start, the one after returns null. |

| ID     | Procedure and pass criterion                                                                                                                                                                                           |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SRV-04 | <b>Start withheld while running.</b> Heartbeat with state <code>running</code> while <code>start</code> is pending. <b>Pass:</b> no job returned and <code>/machines</code> still shows <code>pending:"start"</code> . |
| SRV-05 | <b>Stop always delivered.</b> POST <code>/stop</code> , then heartbeat from a running and an idle host. <b>Pass:</b> both receive <code>stop</code> .                                                                  |
| SRV-06 | <b>Single slot.</b> <code>/start</code> then <code>/stop</code> before the board polls. <b>Pass:</b> only <code>stop</code> is delivered.                                                                              |
| SRV-07 | <b>Pending start after exit.</b> <code>/start</code> while running, then report idle. <b>Pass:</b> <code>start</code> is delivered.                                                                                    |
| SRV-08 | <b>Bad input.</b> Missing host, invalid JSON, no content type, GET <code>/start</code> . <b>Pass:</b> 422, 400, 415, 405, and <code>/machines</code> still answers 200.                                                |
| SRV-09 | <b>Offline exclusion.</b> Stop heartbeating for 16s, then POST <code>/start</code> . <b>Pass:</b> <code>"queued": []</code> .                                                                                          |

Table 4: Server tests.

### 4.3 Agent and board

| ID      | Procedure and pass criterion                                                                                                                                                                                                                                                                                                                                                                         |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INS-01  | <b>Fresh install.</b> Flash a card, run <code>install-software.sh</code> and <code>install-seb.sh</code> with <code>--host</code> , then any manual steps. Record every manual step needed. <b>Pass:</b> <code>systemctl --user is-active seb-agent</code> prints <code>active</code> , <code>~/safe-exam-browser</code> exists and is executable, and the board appears in <code>/machines</code> . |
| AGT-01  | <b>Unique identity.</b> On each board run <code>hostname</code> and <code>systemctl --user show seb-agent -p Environment</code> . <b>Pass:</b> <code>/machines</code> lists each board as a separate host.                                                                                                                                                                                           |
| AGT-02  | <b>Cadence.</b> <code>journalctl --user -u seb-agent -o short-precise -n 25</code> . <b>Pass:</b> intervals match R-T1; acceptable range <i>[TODO]</i> .                                                                                                                                                                                                                                             |
| AGT-03  | <b>Outage tolerance.</b> Make the control URL unreachable for a set time, then restore it. <b>Pass:</b> the agent keeps running, logs each failed heartbeat, and resumes heartbeating after restoration; duration and recovery time <i>[TODO]</i> .                                                                                                                                                  |
| AGT-04  | <b>Agent crash recovery.</b> Kill the agent process. <b>Pass:</b> the systemd unit restarts it; time limit <i>[TODO]</i> .                                                                                                                                                                                                                                                                           |
| AGT-05  | <b>Boot without login.</b> Power-cycle a board and do not log in. <b>Pass:</b> the board is listed in <code>/machines</code> ; time limit <i>[TODO]</i> .                                                                                                                                                                                                                                            |
| SUDO-01 | <b>Sudo scope.</b> As <code>orangepi</code> run <code>sudo -n -l</code> . <b>Pass:</b> only the two scripts are listed.                                                                                                                                                                                                                                                                              |
| SES-01  | <b>Start from console.</b> Run <code>sudo -n ~/start-seb.sh</code> . <b>Pass:</b> SEB is visible on the display; time limit <i>[TODO]</i> .                                                                                                                                                                                                                                                          |
| SES-02  | <b>Stop from console.</b> Run <code>sudo -n ~/stop-seb.sh</code> . <b>Pass:</b> no SEB or <code>xinit</code> process remains, the text console is shown and the keyboard works.                                                                                                                                                                                                                      |
| SES-03  | <b>Repeated cycles.</b> Start and stop repeatedly, recording memory use each cycle. <b>Pass:</b> every cycle meets SES-01 and SES-02; number of cycles and memory tolerance <i>[TODO]</i> .                                                                                                                                                                                                          |
| SES-04  | <b>Duplicate start.</b> While running, send POST <code>/start</code> twice. <b>Pass:</b> a single SEB session.                                                                                                                                                                                                                                                                                       |

| ID     | Procedure and pass criterion                                                                                                                                                            |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SES-05 | <b>Config.</b> Confirm the board has a local start file for the lab launch (not a repo artifact). <b>Pass:</b> SEB starts without a password prompt; expected start URL <i>[TODO]</i> . |

Table 5: Agent and board session tests.

#### 4.4 Integration and demo

| ID      | Procedure and pass criterion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEMO-01 | <b>Two boards start and stop on one command.</b> Pre: SES-01, SES-02 and AGT-01 passed on both boards; both listed <code>idle</code> in <code>/machines</code> . Steps: (1) note the time; (2) <code>curl -sS -X POST \$CTRL/start</code> ; (3) record when SEB appears on each display; (4) <code>GET /machines</code> ; (5) <code>curl -sS -X POST \$CTRL/stop</code> ; (6) record when each display returns to the console. Number of trials <i>[TODO]</i> . <b>Pass:</b> <code>queued</code> lists both boards; SEB runs on both; <code>/machines</code> shows both <code>running</code> ; after stop both are back at the console. Time limits R-T3 to R-T5 <i>[TODO]</i> . |
| DEMO-02 | <b>Board offline at start.</b> Stop one board's agent, send <code>POST /start</code> . <b>Pass:</b> <code>queued</code> lists only the other board. Restart the agent; record whether it launches SEB. Send <code>/start</code> again and record the result.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| DEMO-03 | <b>Board loses network while running.</b> With both running, disconnect one board. Record SEB state, the board's <code>seen_secs_ago</code> , and the <code>queued</code> list for a <code>POST /stop</code> . Reconnect and send <code>/stop</code> again.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| DEMO-04 | <b>Server restart.</b> With both boards running, restart the server. Record how long until both are listed again and with which state.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| EDGE-01 | <b>Pending start after self-exit.</b> While a board reports <code>running</code> , send <code>/start</code> (held), then tear the session down so the agent reports <code>idle</code> . <b>Pass:</b> the next heartbeat receives <code>start</code> and the session relaunches (job rule above).                                                                                                                                                                                                                                                                                                                                                                                 |

Table 6: Integration and demo tests.

#### 4.5 Lockdown and security

| ID     | Procedure and pass criterion                                                                                                                                                                                                                            |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LD-01  | <b>Exit paths.</b> With SEB started by the agent, try the key combinations, mouse actions and window actions the team lists here: <i>[TODO]</i> . <b>Pass:</b> none gives a shell, another application or the console. Record each attempt and outcome. |
| SEC-01 | <b>Secure start.</b> <i>[TODO: write what secure start should mean and how to test it]</i> .                                                                                                                                                            |
| SEC-02 | <b>No inbound surface.</b> On a board run <code>ss -ltnp</code> . <b>Pass:</b> no listening socket belongs to <code>seb-agent</code> .                                                                                                                  |
| SEC-03 | <b>Control-channel access.</b> Record whether <code>POST /start</code> and <code>POST /stop</code> succeed from a machine with no credentials, with no exam running.                                                                                    |

Table 7: Lockdown and security tests.

#### 4.6 Performance and endurance

| ID      | Procedure and pass criterion                                                                                                                                                              |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PERF-01 | <b>Page load.</b> Start SEB with <code>/start</code> ; time the Canvas pages the team lists: <i>[TODO]</i> . Number of runs <i>[TODO]</i> . Report median and maximum. <b>Pass:</b> R-P1. |
| PERF-02 | <b>Input latency.</b> Time click to visible response over a set number of clicks <i>[TODO]</i> . <b>Pass:</b> R-P2.                                                                       |
| PERF-03 | <b>Thermal.</b> Run a Canvas session of set length <i>[TODO]</i> while logging temperature at a set interval <i>[TODO]</i> (log method <i>[TODO]</i> ). <b>Pass:</b> R-P3.                |
| PERF-04 | <b>Endurance.</b> Run SEB for a set duration <i>[TODO]</i> . Count crashes, agent restarts and display glitches. <b>Pass:</b> counts and limits <i>[TODO]</i> .                           |

Table 8: Performance tests.

## 4.7 Demo runbook (DEMO-01)

```
# laptop: start server and tunnel (Docker running)
./scripts/deploy-server.sh

# second terminal
CTRL=https://riscv-exam-computer.download
curl -sS $CTRL/machines # both boards listed, state idle
curl -sS -X POST $CTRL/start # queued lists both boards
curl -sS $CTRL/machines # both boards state running
curl -sS -X POST $CTRL/stop # both boards back at the console
```

## 5 Traceability

| Expected behavior                         | Tests                             |
|-------------------------------------------|-----------------------------------|
| Server routes and job rules               | SRV-01 to SRV-09, EDGE-01         |
| Agent loop, identity, recovery            | AGT-01 to AGT-05, SUDO-01, INS-01 |
| Board session start and stop              | SES-01 to SES-05                  |
| Two-board start and stop (beta objective) | DEMO-01 to DEMO-04                |
| Timing requirements                       | AGT-02, DEMO-01, SES-01, SES-02   |
| Performance requirements                  | PERF-01 to PERF-04                |
| Lockdown and security                     | LD-01, SEC-01 to SEC-03           |
| Alpha findings                            | <i>[TODO]</i>                     |

Table 9: Expected behavior to tests.

## 6 Results record

| Test | Date | Tester | Boards | Result | Measurements and notes |
|------|------|--------|--------|--------|------------------------|
|------|------|--------|--------|--------|------------------------|

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*Table 10: Results record template.*